Completing the square

1 Halve the coefficient of x and then square it.

2

a No

b 9

3 $\frac{9}{4}$

4

a 25

b 1

c 8

d $\frac{1}{4}$

e $\left(\frac{m}{2}\right)^2$

f $\frac{4}{25}$

5

a $x^2 + 4x + 4 = (x+2)^2$

b $x^2 - 5x + \frac{25}{4} = \left(x - \frac{5}{2}\right)^2$

c $x^2 - \frac{7}{4}x + \frac{49}{64} = \left(x - \frac{7}{8}\right)^2$

6

a $(x+6)^2 - 36$

b $(x-6)^2 - 36$

c $(x+1)^2 + 1$

d $(x-7)^2 + 7$

e $(x-3)^2 - 4$

f $\left(x + \frac{3}{2}\right)^2 + \frac{15}{4}$

g $\left(x - \frac{7}{2}\right)^2 + \frac{7}{4}$

7

a $y = (x+3)(x+1)$

b $y = (x-2)(x-4)$

c $y = (x+7)(x+10)$

d $y = (x-4)(x-7)$

e $y = (x+53)(x+3)$

f $y = (x-3)(x-21)$

8 $y = (x+10)(x+2)$

9

a $y = \left(x - 3 + \sqrt{2}\right)\left(x - 3 - \sqrt{2}\right)$

b $y = \left(x + 4 + \sqrt{2}\right)\left(x + 4 - \sqrt{2}\right)$

10

a In step 4, she did not factor the expression, but instead wrote it as the difference of two squares.

b $y = (x-3)(x-35)$

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Non-monic completing the square

11 $4\left(\left(x-\frac{9}{2}\right)^2-\frac{1}{2}\right)$

12

a $3\left(x+\frac{3}{2}\right)^2+\frac{5}{4}$

b $4\left(x-\frac{11}{8}\right)^2-\frac{9}{16}$

13

a $y=(2x+23)(x+1)$

b $y=4(x+9)(x+2)$

Quadratic equations

14

a No **b** Yes **c** No **d** No

e Yes

15

a $x=-4,-8$ **b** $x=\frac{3}{2},\frac{1}{2}$ **c** $x=5,1$ **d** $x=\pm2\sqrt{5}-2$

e $x=-5\pm\sqrt{17}$ **f** $x=-9\pm\frac{\sqrt{30}}{2}$

16

a $x=-2,-16$ **b** $x=4,2$ **c** $x=8,1$ **d** $x=8,-2$

e $x=-1,-\frac{7}{4}$ **f** $x=1\pm\sqrt{33}$

17

a $x=-12\pm\sqrt{139}$

b $x=\frac{-11\pm\sqrt{101}}{2}$

c $x=\pm2\sqrt{3}-4$

d $x=\frac{7}{2}\pm\frac{\sqrt{17}}{2}$

e $x=-\frac{1}{6}\pm\frac{\sqrt{109}}{6}$

f $x=\frac{-11}{2}\pm\frac{\sqrt{2965}}{10}$

Applications

18 $x=3,\ -7$

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19 $8, -8$

20 $x = 12$

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